

Image Segmentation

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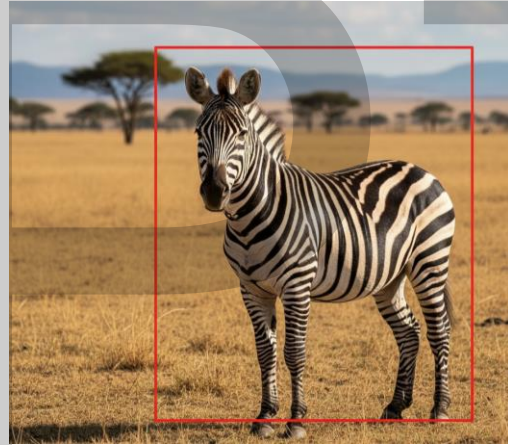
- Introduction to Image Segmentation
- Need for Image segmentation
- Types of Image segmentation
- U-Net Architecture for Image segmentation

Classification, localization and Object Detection

Single Object

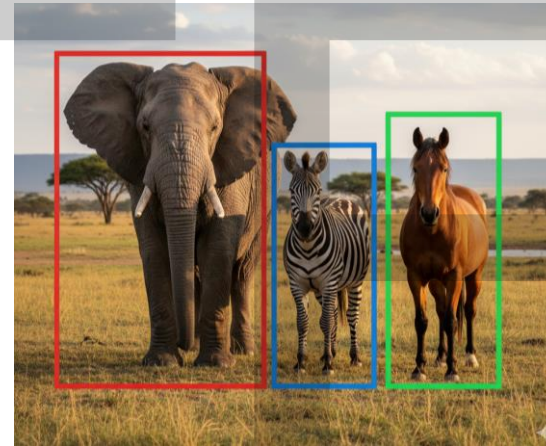


Classification



Classification and Localization

Multiple Objects



Object detection

Introduction to Image Segmentation

Image Segmentation

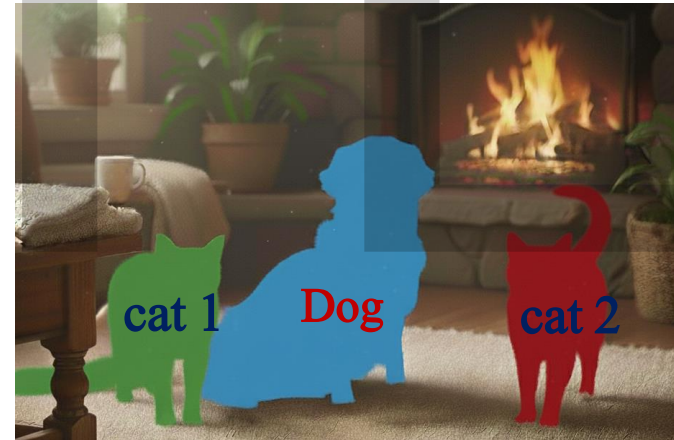
Computer vision technique that divides a digital image into distinct pixel groups, known as image segments, to aid in object detection and similar tasks



Semantic Segmentation



Instance Segmentation



Need for Image Segmentation

Image segmentation helps to extract the object of interest from an image which is further used for processing like recognition and description

Segmentation Mask

A segmentation mask is a specific portion of an image that is isolated from the rest of an image

Segmentation mask is of same size as the original image

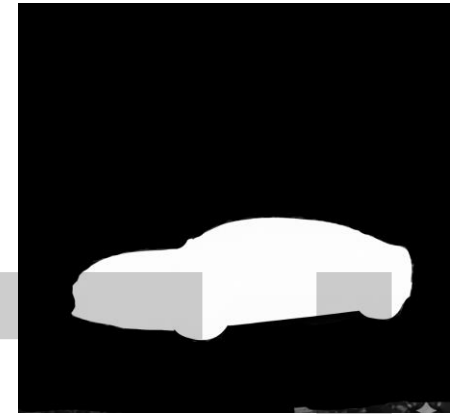
Types of segmentation masks

- ✓ Binary Mask
- ✓ Multiclass mask



Binary Mask

Pixels in the mask are set to a value of 1 (or true) for the region of interest, and 0 (or false) for the background.



Multiclass Mask

Multiclass mask can contain a unique value for each class, such as 0 for the background, 1 for dog and 2 for cat.

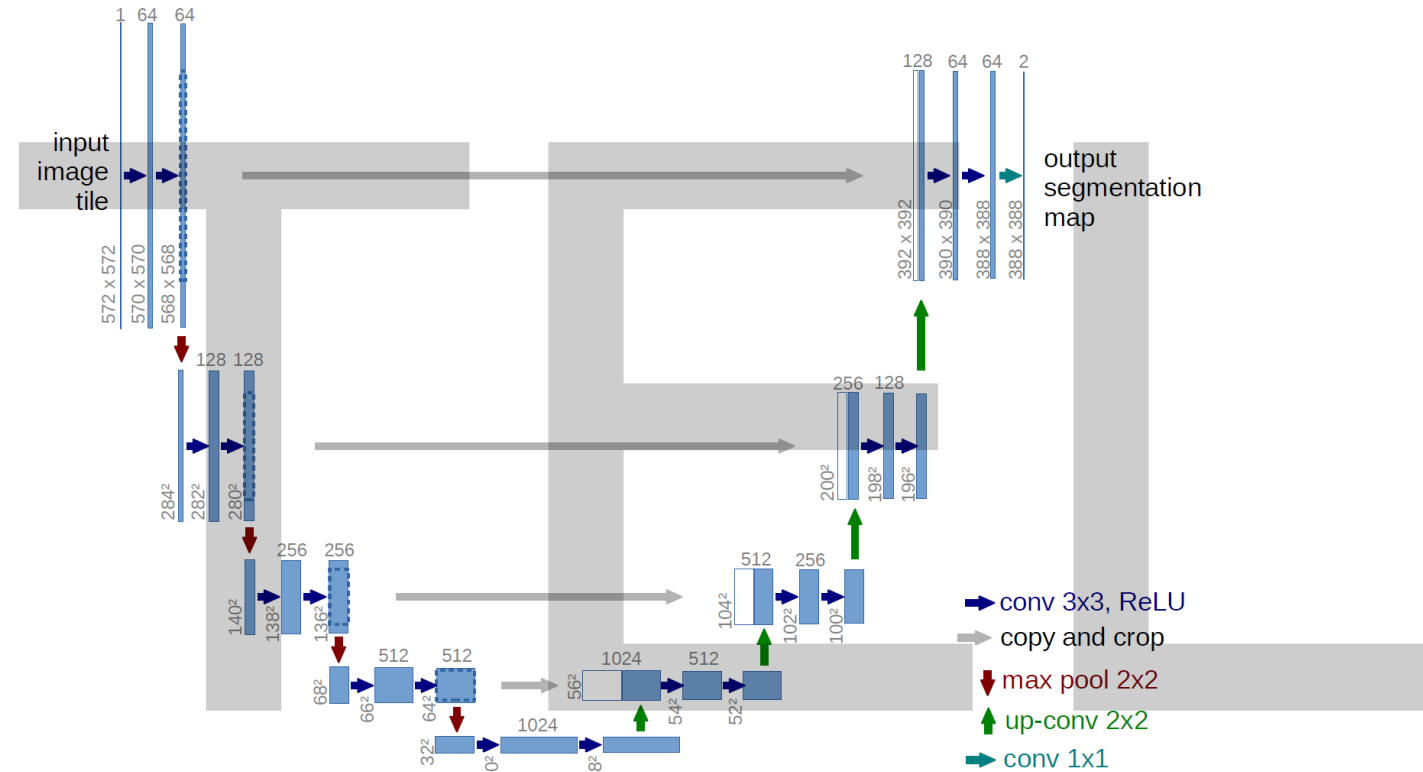
Deep Learning Algorithm for Image Segmentation

U-Net Architecture

Introduced in 2015 for biomedical image segmentation in the paper

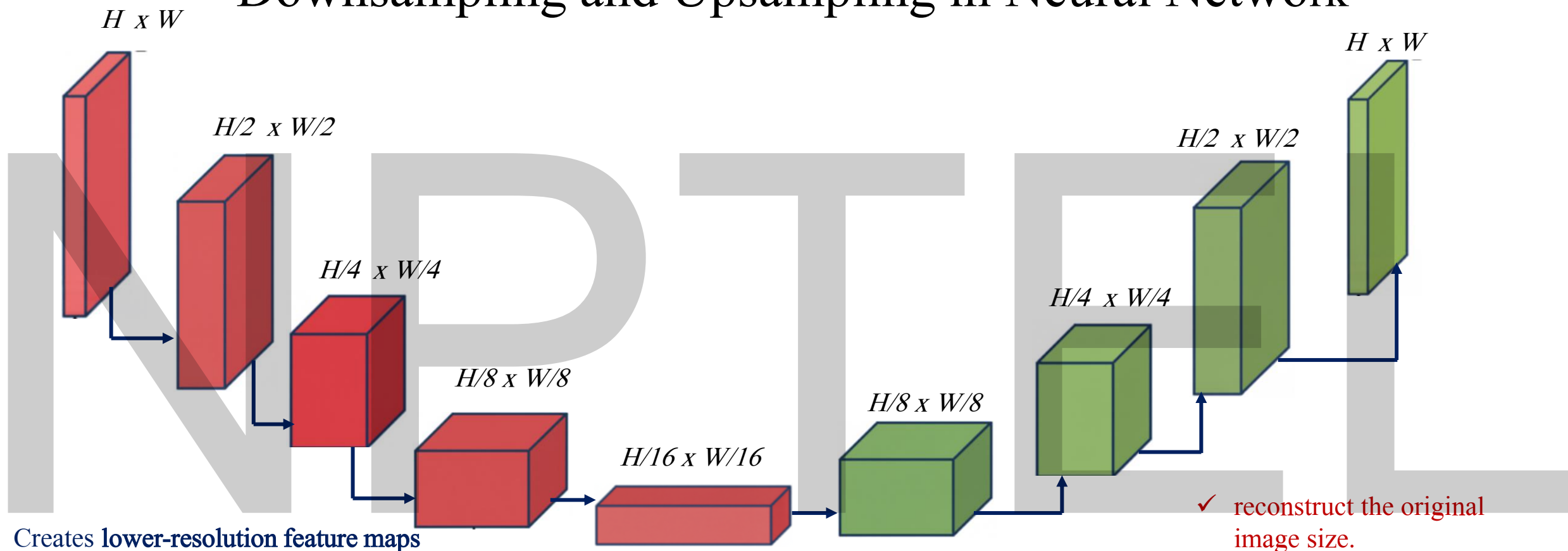
“*U-Net: Convolutional Networks for Biomedical Image Segmentation*”

by Olaf Ronneberger, Philipp Fischer, and Thomas Brox



Ref:
Olaf Ronneberger, Philipp Fischer, and Thomas Brox
U-Net: Convolutional Networks for Biomedical Image Segmentation
[arXiv:1505.04597v1](https://arxiv.org/abs/1505.04597v1) [cs.CV]

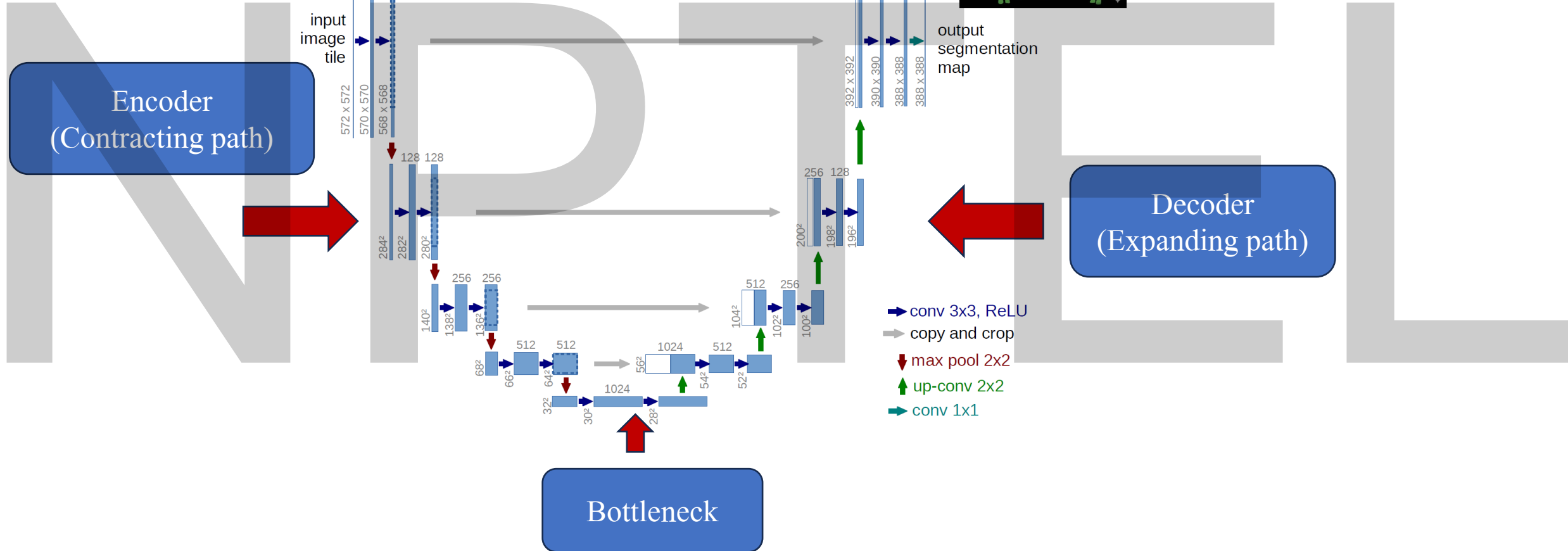
Downsampling and Upsampling in Neural Network



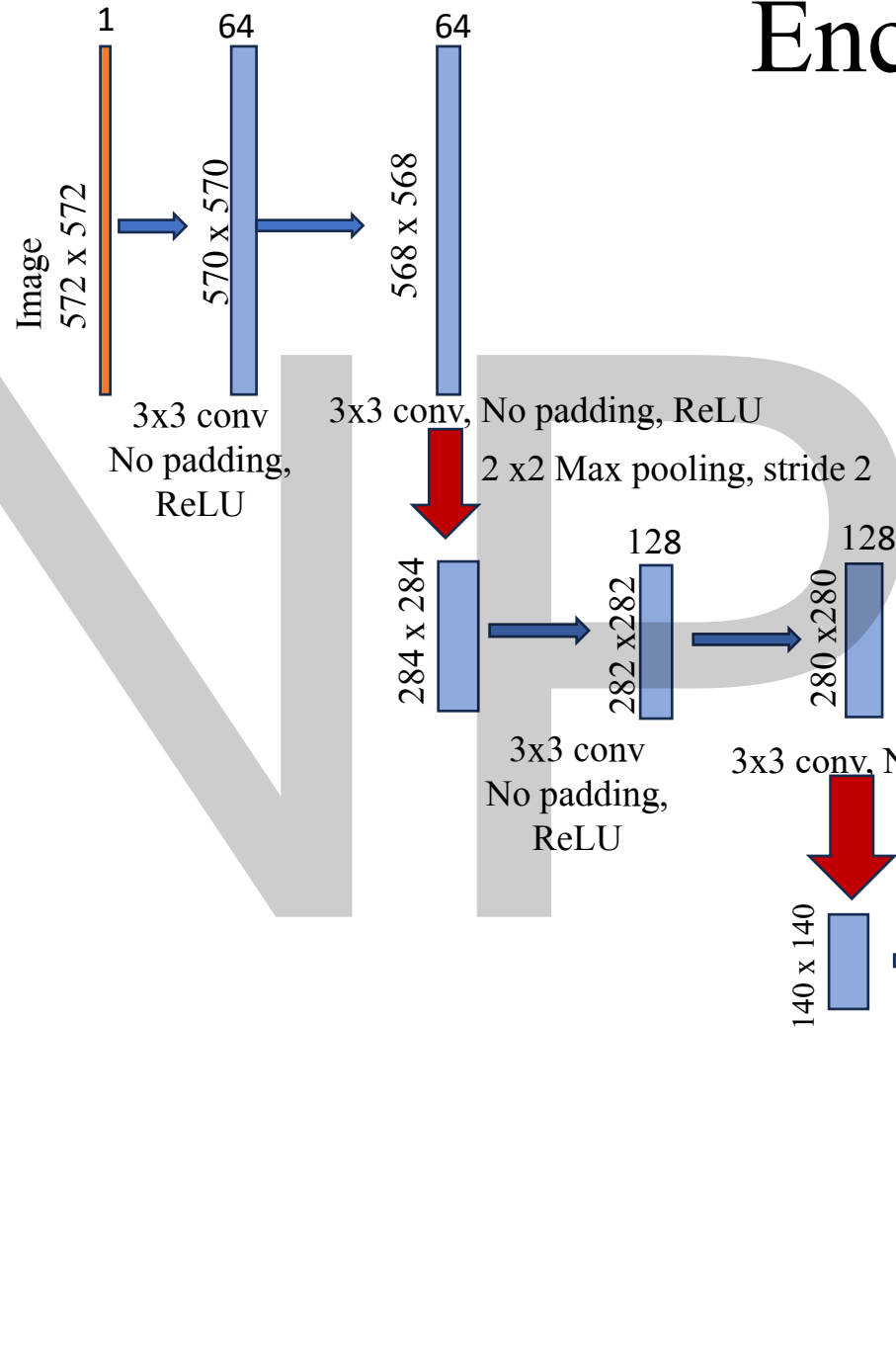
- ✓ Creates lower-resolution feature maps that focus on *what* is in the image rather than fine details of it.
- ✓ Reduces the dimensionality of the data thus enabling in faster processing of the data (image).

- ✓ Required for making a pixel-by-pixel segmentation mask that matches the original image dimension

U-Net Architecture



Encoder Block



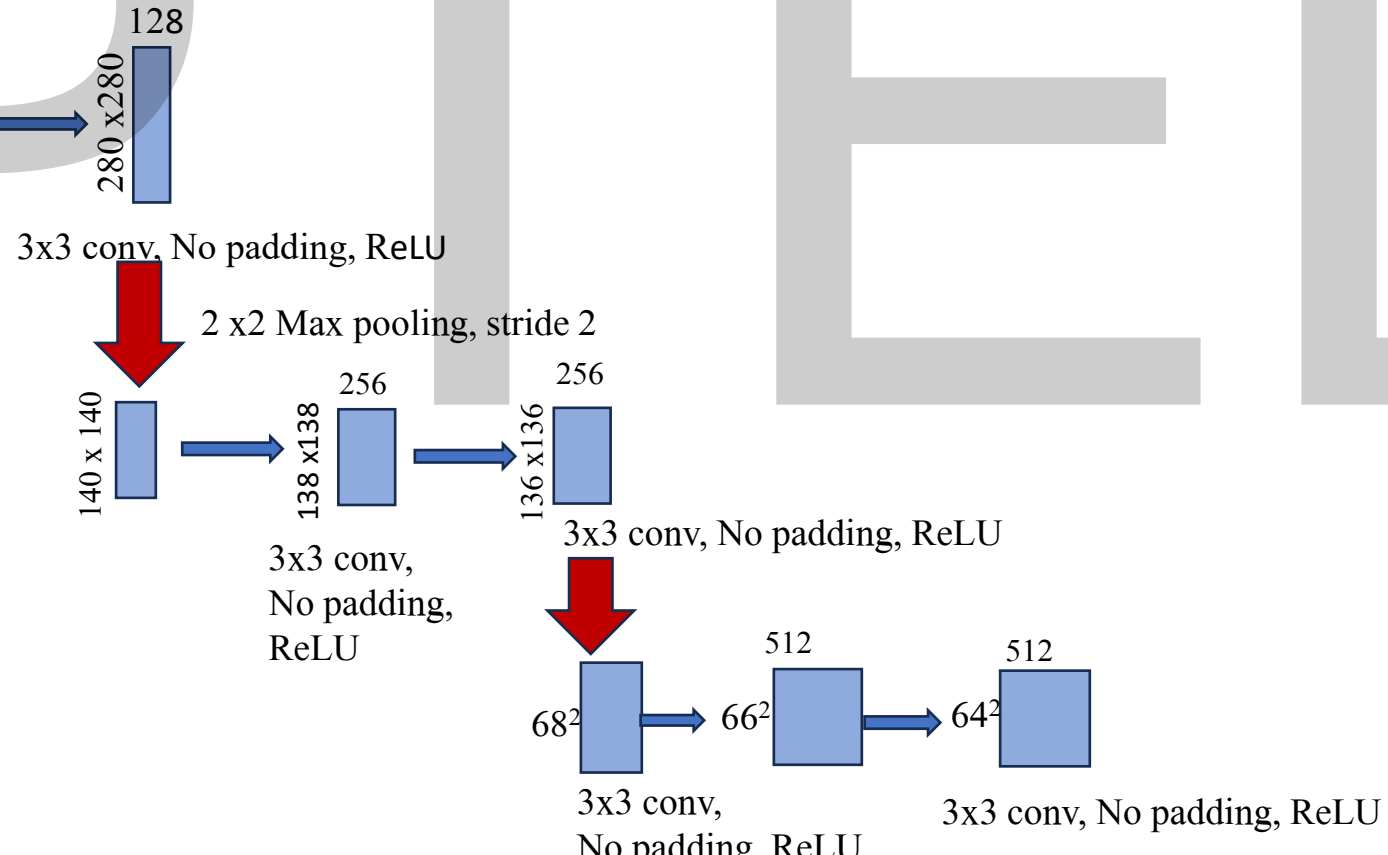
✓ Why do we need the encoder?

It acts as the feature extractor.

It shrinks the image size while keeping important information

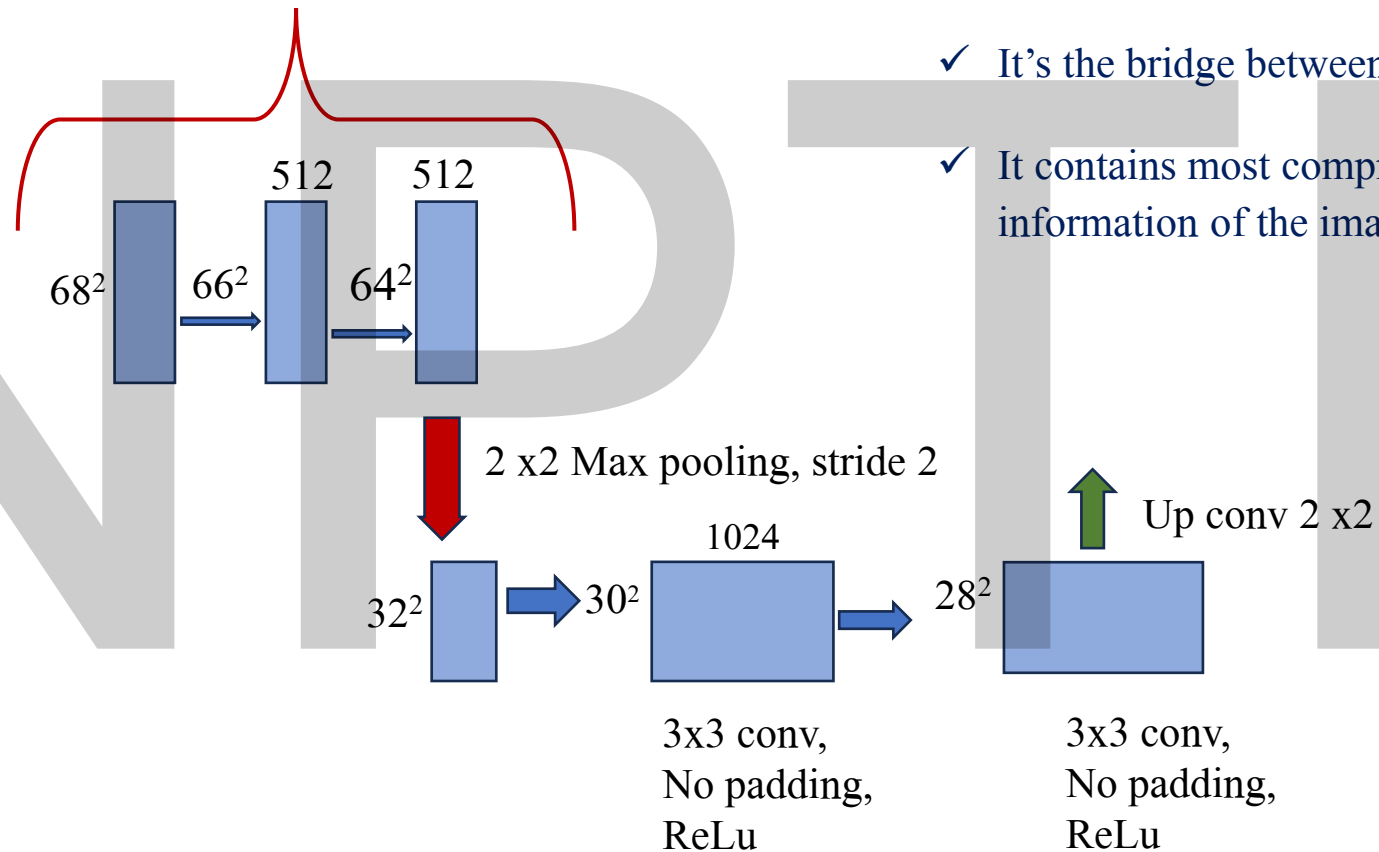
✓ Why the number of filters are increasing with every step?

More number of channels allow the model to capture higher-level patterns.



Bottleneck (Bottom of the U)

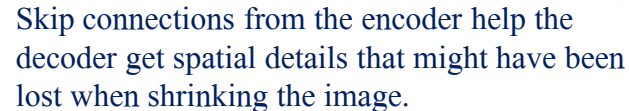
Last block of the encoder



✓ What is the need for the bottleneck layer?

✓ It's the bridge between the encoder and the decoder

✓ It contains most compressed, abstract and meaningful information of the image.



Significance of various blocks in U-Net Architecture

Encoder block

- Extracts features and reduces spatial size (downsampling).
- Helps the network learn contextual information and abstract features.

Decoder block

- The decoder reconstructs the segmentation map by upsampling the compressed representation from the encoder.
- Helps regain spatial details lost during encoding.
- Helps in precise localization of structures.

Bottle Neck

- Bridges the encoder (contracting path) and the decoder (expanding path).
- Acts as the most compressed representation of the input image,
- Captures highly abstracted features before the decoder begins upsampling

Skip Connection

- Preserve spatial detail that would otherwise be lost in down sampling.
- Feature maps from encoder layers are copied and concatenated with corresponding decoder layers

NPTFEL

Thank you